SG No. 224 O_{h}^{4} $Pn\bar{3}m$ [cubic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `2a`, site symmetry: `-43m`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,-3,4,-41,42,43,-44]`
2	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6,-7,8,-37,38,39,-40]`
3	$[0,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10,-11,12,45,-46,-47,48]`
4	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,-14,-15,16,-29,30,31,-32]`
5	$[X,\ 0,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[17,-18,-19,20,-25,26,27,-28]`
6	$[0,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[21,-22,-23,24,33,-34,-35,36]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,5,9,38,43,48]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,7,12,37,41,45]`
3	$[-X,\ X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,8,10,39,44,46]`
4	$[X,\ -X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,6,11,40,42,47]`
5	$[X,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,17,21,26,31,36]`
6	$[-X,\ -X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,19,24,25,29,33]`
7	$[X,\ -X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15,20,22,27,32,34]`
8	$[-X,\ X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[16,18,23,28,30,35]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,-37,38]`
2	$[-X,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,-4,39,-40]`
3	$[0,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6,-47,48]`
4	$[0,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,-8,45,-46]`
5	$[X,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10,43,-44]`
6	$[X,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[11,-12,-41,42]`
7	$[X,\ X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,-14,-25,26]`
8	$[X,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15,-16,27,-28]`
9	$[X,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[17,-18,-35,36]`
10	$[-X,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[19,-20,33,-34]`
11	$[0,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[21,-22,31,-32]`
12	$[0,\ X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[23,-24,-29,30]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\ X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,38]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[2,37]
3	$[-X, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[3,39]
4	$[X, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[4,40]
5	$[Y, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,48]
6	$[Y, -X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[6,47]
7	$[-Y, -X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[7,45]
8	$[-Y, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[8,46]
9	$[X, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9,43]
10	$[-X, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[10,44]
11	$[X, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[11,42]
12	$[-X, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[12,41]
13	$[X, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[13,26]
14	$[-X, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[14,25]
15	$[X, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[15,27]
16	$[-X, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[16,28]
17	$[X, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[17,36]
18	$[-X, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[18,35]
19	$[-X, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[19,33]
20	$[X, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[20,34]
21	$[Y, X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[21,31]
22	$[Y, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[22,32]
23	$[-Y, X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[23,30]
24	$[-Y, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[24,29]

Table 5: Wyckoff bond: 24e@2a, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-2]
2	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[3,-4]
3	$[0, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,-6]
4	$[0, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[7,-8]
5	$[Y, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9,-10]
6	$[Y, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[11,-12]
7	$[Y, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[13,-14]
8	$[Y, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[15,-16]
9	$[X, 0, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[17,-18]
10	$[-X, 0, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[19,-20]
11	$[0, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[21,-22]
12	$[0, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[23,-24]
13	$[-X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,-26]
14	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[27,-28]
15	$[0, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[29,-30]
16	$[0, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[31,-32]
17	$[-Y, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[33,-34]
18	$[-Y, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[35,-36]

continued ...

Table 5

No.	vector	center	mapping
19	$[-Y, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[37,-38]
20	$[-Y, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[39,-40]
21	$[-X, 0, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[41,-42]
22	$[X, 0, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[43,-44]
23	$[0, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[45,-46]
24	$[0, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[47,-48]

Table 6: Wyckoff bond: 48f@2a, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1]
2	$[-X, -Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[2]
3	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[3]
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[4]
5	$[Z, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5]
6	$[Z, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[6]
7	$[-Z, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[7]
8	$[-Z, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[8]
9	$[Y, Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9]
10	$[-Y, Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[10]
11	$[Y, -Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[11]
12	$[-Y, -Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[12]
13	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[13]
14	$[-Y, -X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[14]
15	$[Y, -X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[15]
16	$[-Y, X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[16]
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[17]
18	$[-X, Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[18]
19	$[-X, -Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[19]
20	$[X, -Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[20]
21	$[Z, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[21]
22	$[Z, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[22]
23	$[-Z, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[23]
24	$[-Z, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[24]
25	$[-X, -Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[25]
26	$[X, Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[26]
27	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[27]
28	$[-X, Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[28]
29	$[-Z, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[29]
30	$[-Z, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[30]
31	$[Z, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[31]
32	$[Z, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[32]
33	$[-Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[33]
34	$[Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[34]
35	$[-Y, Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[35]

continued ...

Table 6

No.	vector	center	mapping
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[48]

* Wyckoff site: 4b, site symmetry: .-3m

Table 7: Wyckoff bond: 4a@4b, bond symmetry: .-3m

No.	vector	center	mapping
1	$[X, X, X]$	$[0, 0, 0]$	[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]
2	$[-X, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]
3	$[-X, X, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]
4	$[X, -X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]

Table 8: Wyckoff bond: 12b@4b, bond symmetry: ..2/m

No.	vector	center	mapping
1	$[X, X, Y]$	$[0, 0, 0]$	[1,-14,-25,38]
2	$[-X, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,-13,-26,37]
3	$[-X, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,-15,-27,39]
4	$[X, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,-16,-28,40]
5	$[Y, X, X]$	$[0, 0, 0]$	[5,-24,-29,48]
6	$[Y, -X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,-23,-30,47]
7	$[-Y, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[7,-21,-31,45]
8	$[-Y, X, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[8,-22,-32,46]
9	$[X, Y, X]$	$[0, 0, 0]$	[9,-19,-33,43]
10	$[-X, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[10,-20,-34,44]
11	$[X, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[11,-18,-35,42]
12	$[-X, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[12,-17,-36,41]

Table 9: Wyckoff bond: 12c@4b, bond symmetry: ..2/m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, 0, 0]$	[1,14,-25,-38]

continued ...

Table 9

No.	vector	center	mapping
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,13,-26,-37]`
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,15,-27,-39]`
4	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,16,-28,-40]`
5	$[0, X, -X]$	$[0, 0, 0]$	`[5,24,-29,-48]`
6	$[0, -X, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,23,-30,-47]`
7	$[0, -X, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,21,-31,-45]`
8	$[0, X, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,22,-32,-46]`
9	$[-X, 0, X]$	$[0, 0, 0]$	`[9,19,-33,-43]`
10	$[X, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,20,-34,-44]`
11	$[-X, 0, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,18,-35,-42]`
12	$[X, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,17,-36,-41]`

Table 10: Wyckoff bond: `24d@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$[0, 0, 0]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$[0, 0, 0]$	`[24,-48]`

* Wyckoff site: `4c`, site symmetry: `.-3m`

Table 11: Wyckoff bond: `4a@4c`, bond symmetry: `.-3m`

No.	vector	center	mapping
1	$[X, X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]`
2	$[-X, -X, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]`
3	$[-X, X, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]`
4	$[X, -X, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]`

Table 12: Wyckoff bond: `12b@4c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-14,-25,38]`
2	$[-X, -X, Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-13,-26,37]`
3	$[-X, X, -Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-15,-27,39]`
4	$[X, -X, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-16,-28,40]`
5	$[Y, X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-24,-29,48]`
6	$[Y, -X, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-23,-30,47]`
7	$[-Y, -X, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-21,-31,45]`
8	$[-Y, X, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[8,-22,-32,46]`
9	$[X, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-19,-33,43]`
10	$[-X, Y, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[10,-20,-34,44]`
11	$[X, -Y, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11,-18,-35,42]`
12	$[-X, -Y, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[12,-17,-36,41]`

Table 13: Wyckoff bond: `12c@4c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,14,-25,-38]`
2	$[-X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,13,-26,-37]`
3	$[-X, -X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,15,-27,-39]`
4	$[X, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,16,-28,-40]`
5	$[0, X, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,24,-29,-48]`
6	$[0, -X, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,23,-30,-47]`
7	$[0, -X, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,21,-31,-45]`
8	$[0, X, X]$	$\left[0, \frac{1}{2}, 0\right]$	`[8,22,-32,-46]`
9	$[-X, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,19,-33,-43]`
10	$[X, 0, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[10,20,-34,-44]`
11	$[-X, 0, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11,18,-35,-42]`
12	$[X, 0, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[12,17,-36,-41]`

Table 14: Wyckoff bond: `24d@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-27]`

continued ...

Table 14

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-28]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[8,-32]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[0, \frac{1}{2}, 0\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[24,-48]`

* Wyckoff site: `6d`, site symmetry: `-42.m`

Table 15: Wyckoff bond: `6a@6d`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,-2,-3,4,-41,42,43,-44]`
2	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,-6,-7,8,-37,38,39,-40]`
3	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,-10,-11,12,45,-46,-47,48]`
4	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,-14,-15,16,-29,30,31,-32]`
5	$[X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[17,-18,-19,20,-25,26,27,-28]`
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[21,-22,-23,24,33,-34,-35,36]`

Table 16: Wyckoff bond: `12b@6d`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,-4,-42,43]`
2	$[0, -X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,-3,41,-44]`
3	$[X, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,-8,38,-39]`
4	$[X, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,-7,-37,40]`
5	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,-12,-45,48]`
6	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,-11,46,-47]`
7	$[X, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,-16,-30,31]`
8	$[-X, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[14,-15,29,-32]`
9	$[0, X, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[17,-20,26,-27]`

continued ...

Table 16

No.	vector	center	mapping
10	$[0, X, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[18,-19,-25,28]`
11	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[21,-24,-33,36]`
12	$[X, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[22,-23,34,-35]`

Table 17: Wyckoff bond: `12c@6d`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,2,-3,-4]`
2	$[X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,6,-7,-8]`
3	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,10,-11,-12]`
4	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,14,-15,-16]`
5	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[17,18,-19,-20]`
6	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[21,22,-23,-24]`
7	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[25,26,-27,-28]`
8	$[-X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[29,30,-31,-32]`
9	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[33,34,-35,-36]`
10	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[37,38,-39,-40]`
11	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[41,42,-43,-44]`
12	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[45,46,-47,-48]`

Table 18: Wyckoff bond: `24d@6d`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,43]`
2	$[-Y, -X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,41]`
3	$[-Y, X, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,44]`
4	$[Y, -X, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,42]`
5	$[X, Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,38]`
6	$[X, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,40]`
7	$[-X, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,37]`
8	$[-X, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[8,39]`
9	$[X, X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,48]`
10	$[-X, X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,46]`
11	$[X, -X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[11,47]`
12	$[-X, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[12,45]`
13	$[X, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,31]`
14	$[-X, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[14,29]`
15	$[X, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[15,32]`
16	$[-X, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[16,30]`
17	$[Y, X, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[17,26]`
18	$[-Y, X, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[18,28]`
19	$[-Y, -X, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[19,25]`
20	$[Y, -X, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[20,27]`
21	$[X, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[21,36]`

continued ...

Table 18

No.	vector	center	mapping
22	$[X, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[22,34]
23	$[-X, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[23,35]
24	$[-X, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[24,33]

Table 19: Wyckoff bond: 24e@6d, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[1,-2]
2	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[3,-4]
3	$[0, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[5,-6]
4	$[0, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[7,-8]
5	$[Y, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[9,-10]
6	$[Y, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[11,-12]
7	$[Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[13,-14]
8	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[15,-16]
9	$[X, 0, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[17,-18]
10	$[-X, 0, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[19,-20]
11	$[0, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[21,-22]
12	$[0, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[23,-24]
13	$[-X, -Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[25,-26]
14	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[27,-28]
15	$[0, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[29,-30]
16	$[0, X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[31,-32]
17	$[-Y, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[33,-34]
18	$[-Y, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[35,-36]
19	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[37,-38]
20	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[39,-40]
21	$[-X, 0, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[41,-42]
22	$[X, 0, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[43,-44]
23	$[0, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[45,-46]
24	$[0, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[47,-48]

Table 20: Wyckoff bond: 24f@6d, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[1,-4]
2	$[0, -X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[2,-3]
3	$[Y, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[5,-8]
4	$[Y, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[6,-7]
5	$[X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[9,-12]
6	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[10,-11]
7	$[X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[13,-16]
8	$[-X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[14,-15]
9	$[0, Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[17,-20]

continued ...

Table 20

No.	vector	center	mapping
10	$[0, Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[18,-19]`
11	$[Y, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,-24]`
12	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22,-23]`
13	$[0, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[25,-28]`
14	$[0, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[26,-27]`
15	$[-Y, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[29,-32]`
16	$[-Y, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[30,-31]`
17	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[33,-36]`
18	$[X, -Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[34,-35]`
19	$[-X, 0, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[37,-40]`
20	$[X, 0, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[38,-39]`
21	$[0, -Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[41,-44]`
22	$[0, -Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[42,-43]`
23	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[45,-48]`
24	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[46,-47]`

Table 21: Wyckoff bond: `48g@6d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[24]`
25	$[-X, -Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[25]`
26	$[X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[26]`

continued ...

Table 21

No.	vector	center	mapping
27	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[48]

* Wyckoff site: 8e, site symmetry: .3m

Table 22: Wyckoff bond: 8a@8e, bond symmetry: .3m

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9,38,43,48]
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[2,7,12,37,41,45]
3	$[-X, X, -X]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[3,8,10,39,44,46]
4	$[X, -X, -X]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[4,6,11,40,42,47]
5	$[X, X, -X]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, -x\right]$	[13,17,21,26,31,36]
6	$[-X, -X, -X]$	$[-x, -x, -x]$	[14,19,24,25,29,33]
7	$[X, -X, X]$	$\left[x + \frac{1}{2}, -x, x + \frac{1}{2}\right]$	[15,20,22,27,32,34]
8	$[-X, X, X]$	$\left[-x, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[16,18,23,28,30,35]

Table 23: Wyckoff bond: 24b@8e, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, x]$	[1,38]
2	$[-X, -X, Y]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[2,37]
3	$[-X, X, -Y]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[3,39]
4	$[X, -X, -Y]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[4,40]
5	$[Y, X, X]$	$[x, x, x]$	[5,48]
6	$[Y, -X, -X]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[6,47]

continued ...

Table 23

No.	vector	center	mapping
7	$[-Y, -X, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[7,45]
8	$[-Y, X, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[8,46]
9	$[X, Y, X]$	$[x, x, x]$	[9,43]
10	$[-X, Y, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[10,44]
11	$[X, -Y, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[11,42]
12	$[-X, -Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[12,41]
13	$[X, X, -Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[13,26]
14	$[-X, -X, -Y]$	$[-x, -x, -x]$	[14,25]
15	$[X, -X, Y]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[15,27]
16	$[-X, X, Y]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[16,28]
17	$[X, Y, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[17,36]
18	$[-X, Y, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[18,35]
19	$[-X, -Y, -X]$	$[-x, -x, -x]$	[19,33]
20	$[X, -Y, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[20,34]
21	$[Y, X, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[21,31]
22	$[Y, -X, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[22,32]
23	$[-Y, X, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[23,30]
24	$[-Y, -X, -X]$	$[-x, -x, -x]$	[24,29]

Table 24: Wyckoff bond: 24c@8e, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, x]$	[1,-38]
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[2,-37]
3	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[3,-39]
4	$[X, X, 0]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[4,-40]
5	$[0, X, -X]$	$[x, x, x]$	[5,-48]
6	$[0, -X, X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[6,-47]
7	$[0, -X, -X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[7,-45]
8	$[0, X, X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[8,-46]
9	$[-X, 0, X]$	$[x, x, x]$	[9,-43]
10	$[X, 0, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[10,-44]
11	$[-X, 0, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[11,-42]
12	$[X, 0, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[12,-41]
13	$[-X, X, 0]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[13,-26]
14	$[X, -X, 0]$	$[-x, -x, -x]$	[14,-25]
15	$[-X, -X, 0]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[15,-27]
16	$[X, X, 0]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[16,-28]
17	$[X, 0, X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[17,-36]
18	$[-X, 0, -X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[18,-35]
19	$[-X, 0, X]$	$[-x, -x, -x]$	[19,-33]
20	$[X, 0, -X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[20,-34]
21	$[0, -X, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[21,-31]
22	$[0, X, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[22,-32]
23	$[0, -X, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[23,-30]
24	$[0, X, -X]$	$[-x, -x, -x]$	[24,-29]

Table 25: Wyckoff bond: `48d@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[13]
14	$[-Y, -X, -Z]$	$[-x, -x, -x]$	[14]
15	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[17]
18	$[-X, Z, Y]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -x, -x]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[21]
22	$[Z, -Y, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$[-x, -x, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[29]
30	$[-Z, X, Y]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[31]
32	$[Z, -X, Y]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[33]
34	$[Y, -Z, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[37]
38	$[Y, X, Z]$	$[x, x, x]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[39]
40	$[Y, -X, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[42]
43	$[X, Z, Y]$	$[x, x, x]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[46]

continued ...

Table 25

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[47]`
48	$[Z, Y, X]$	$[x, x, x]$	`[48]`

* Wyckoff site: `12f`, site symmetry: `2.22`

Table 26: Wyckoff bond: `12a@12f`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-4,18,-19]`
2	$[0, -X, X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-3,-17,20]`
3	$[X, 0, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-11,-22,23]`
7	$[0, -X, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,-28,42,-43]`
8	$[0, X, -X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,-27,-41,44]`
9	$[-X, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,-32,-38,39]`
10	$[-X, 0, X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,-31,37,-40]`
11	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,-36,45,-48]`
12	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,-35,-46,47]`

Table 27: Wyckoff bond: `12b@12f`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-4,-18,19]`
2	$[0, -X, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-3,17,-20]`
3	$[-X, 0, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-8,14,-15]`
4	$[-X, 0, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,-7,-13,16]`
5	$[X, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,-12,-21,24]`
6	$[-X, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-11,22,-23]`
7	$[0, -X, X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,-28,-42,43]`
8	$[0, X, X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,-27,41,-44]`
9	$[X, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,-32,38,-39]`
10	$[X, 0, X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,-31,-37,40]`
11	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,-36,-45,48]`
12	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,-35,46,-47]`

Table 28: Wyckoff bond: `12c@12f`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,-18,-19]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,-17,-20]`
3	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,8,-14,-15]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,7,-13,-16]`

continued ...

Table 28

No.	vector	center	mapping
5	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,12,-21,-24]`
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,11,-22,-23]`
7	$[-X, 0, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,28,-42,-43]`
8	$[X, 0, 0]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,27,-41,-44]`
9	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,32,-38,-39]`
10	$[0, X, 0]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,31,-37,-40]`
11	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,36,-45,-48]`
12	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,35,-46,-47]`

Table 29: Wyckoff bond: `24d@12f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-19]`
2	$[-Y, -X, X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,-21]`
13	$[-Y, -X, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,-43]`
14	$[Y, X, -X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,-41]`
15	$[Y, -X, X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[27,-44]`
16	$[-Y, X, X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[28,-42]`
17	$[-X, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,-38]`
18	$[-X, Y, X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,-40]`
19	$[X, Y, -X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[31,-37]`
20	$[X, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[32,-39]`
21	$[-X, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,-48]`
22	$[X, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,-46]`
23	$[-X, X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[35,-47]`
24	$[X, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[36,-45]`

Table 30: Wyckoff bond: `24e@12f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-18]`
2	$[-Y, -X, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-20]`
3	$[-Y, X, X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-17]`
4	$[Y, -X, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-19]`

continued ...

Table 30

No.	vector	center	mapping
5	$[-X, Y, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-15]`
6	$[-X, -Y, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,-13]`
7	$[X, -Y, X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[7,-16]`
8	$[X, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[8,-14]`
9	$[X, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,-21]`
10	$[-X, -X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-23]`
11	$[X, X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[11,-22]`
12	$[-X, X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,-24]`
13	$[-Y, -X, X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,-42]`
14	$[Y, X, X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,-44]`
15	$[Y, -X, -X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[27,-41]`
16	$[-Y, X, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[28,-43]`
17	$[X, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,-39]`
18	$[X, Y, X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,-37]`
19	$[-X, Y, -X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[31,-40]`
20	$[-X, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[32,-38]`
21	$[-X, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,-45]`
22	$[X, X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,-47]`
23	$[-X, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[35,-46]`
24	$[X, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[36,-48]`

Table 31: Wyckoff bond: `24f@12f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[22,-23]`
13	$[0, -X, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25,-28]`
14	$[0, X, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26,-27]`
15	$[-Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29,-32]`
16	$[-Y, 0, X]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30,-31]`
17	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33,-36]`
18	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34,-35]`
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[37,-40]`
20	$[X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[38,-39]`
21	$[0, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[41,-44]`

continued ...

Table 31

No.	vector	center	mapping
22	$[0, -Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[42,-43]`
23	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[45,-48]`
24	$[-Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[46,-47]`

Table 32: Wyckoff bond: `48g@12f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[24]`
25	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[25]`
26	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[26]`
27	$[X, -Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[27]`
28	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[28]`
29	$[-Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[29]`
30	$[-Z, X, Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[30]`
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[31]`
32	$[Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[33]`
34	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[38]`

continued ...

Table 32

No.	vector	center	mapping
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[48]

* Wyckoff site: 12g, site symmetry: 2.mm

Table 33: Wyckoff bond: 12a@12g, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[0, X, X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,-4,-42,43]
2	$[0, -X, X]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	[2,-3,41,-44]
3	$[X, 0, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,-8,38,-39]
4	$[X, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[6,-7,-37,40]
5	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9,-12,-45,48]
6	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[10,-11,46,-47]
7	$[X, 0, -X]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[13,-16,-30,31]
8	$[-X, 0, -X]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[14,-15,29,-32]
9	$[0, X, -X]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[17,-20,26,-27]
10	$[0, X, X]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[18,-19,-25,28]
11	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[21,-24,-33,36]
12	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[22,-23,34,-35]

Table 34: Wyckoff bond: 12b@12g, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,4,42,43]
2	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	[2,3,41,44]
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,8,38,39]
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[6,7,37,40]
5	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9,12,45,48]
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[10,11,46,47]
7	$[0, X, 0]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[13,16,30,31]
8	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[14,15,29,32]
9	$[X, 0, 0]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[17,20,26,27]
10	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[18,19,25,28]
11	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[21,24,33,36]
12	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[22,23,34,35]

Table 35: Wyckoff bond: `24c@12g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,43]`
2	$[-Y, -X, X]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	`[2,41]`
3	$[-Y, X, -X]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	`[3,44]`
4	$[Y, -X, -X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[4,42]`
5	$[X, Y, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,38]`
6	$[X, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	`[6,40]`
7	$[-X, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	`[7,37]`
8	$[-X, Y, -X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[8,39]`
9	$[X, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,48]`
10	$[-X, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	`[10,46]`
11	$[X, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	`[11,47]`
12	$[-X, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[12,45]`
13	$[X, Y, -X]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	`[13,31]`
14	$[-X, -Y, -X]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[14,29]`
15	$[X, -Y, X]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[15,32]`
16	$[-X, Y, X]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	`[16,30]`
17	$[Y, X, -X]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[17,26]`
18	$[-Y, X, X]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[18,28]`
19	$[-Y, -X, -X]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[19,25]`
20	$[Y, -X, X]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[20,27]`
21	$[X, X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[21,36]`
22	$[X, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	`[22,34]`
23	$[-X, X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	`[23,35]`
24	$[-X, -X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[24,33]`

Table 36: Wyckoff bond: `24d@12g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	`[22,-23]`
13	$[0, -X, -Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[25,-28]`
14	$[0, X, -Y]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[26,-27]`
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[29,-32]`
16	$[-Y, 0, X]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	`[30,-31]`
17	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[33,-36]`

continued ...

Table 36

No.	vector	center	mapping
18	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[46,-47]

Table 37: Wyckoff bond: 48e@12g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34]

continued ...

Table 37

No.	vector	center	mapping
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[48]

* Wyckoff site: 24h, site symmetry: 2..

Table 38: Wyckoff bond: 24a@24h, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[17,-20]
10	$[0, Y, X]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[46,-47]

Table 39: Wyckoff bond: `24b@24h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3]`
3	$[0, X, 0]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[6,7]`
5	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[9,12]`
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[10,11]`
7	$[0, X, 0]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[13,16]`
8	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	`[14,15]`
9	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[17,20]`
10	$[-X, 0, 0]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	`[18,19]`
11	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	`[21,24]`
12	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	`[22,23]`
13	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[25,28]`
14	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[26,27]`
15	$[0, -X, 0]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[29,32]`
16	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[30,31]`
17	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	`[33,36]`
18	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	`[34,35]`
19	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[37,40]`
20	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[38,39]`
21	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[41,44]`
22	$[X, 0, 0]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	`[42,43]`
23	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	`[45,48]`
24	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[46,47]`

Table 40: Wyckoff bond: `48c@24h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[8]`
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[17]`

continued ...

Table 40

No.	vector	center	mapping
18	$[-X, Z, Y]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[48]

* Wyckoff site: 24i, site symmetry: ..2

Table 41: Wyckoff bond: 24a@24i, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{2}, y, y+\frac{1}{2}\right]$	[1,-18]
2	$[-Y, -X, -X]$	$\left[0, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[2,-20]
3	$[-Y, X, X]$	$[0, y, -y]$	[3,-17]
4	$[Y, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}-y, -y\right]$	[4,-19]
5	$[-X, Y, X]$	$\left[y+\frac{1}{2}, \frac{1}{2}, y\right]$	[5,-15]
6	$[-X, -Y, -X]$	$\left[y+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	[6,-13]
7	$[X, -Y, X]$	$[-y, 0, y]$	[7,-16]
8	$[X, Y, -X]$	$\left[-y, \frac{1}{2}, \frac{1}{2}-y\right]$	[8,-14]
9	$[X, -X, Y]$	$\left[y, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,-21]
10	$[-X, -X, -Y]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, 0\right]$	[10,-23]

continued ...

Table 41

No.	vector	center	mapping
11	$[X,\ X,\ -Y]$	$[y,\ -y,\ 0]$	`[11,-22]`
12	$[-X,\ X,\ Y]$	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{2}\right]$	`[12,-24]`
13	$[-Y,\ -X,\ X]$	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-y\right]$	`[25,-42]`
14	$[Y,\ X,\ X]$	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[26,-44]`
15	$[Y,\ -X,\ -X]$	$[0,\ -y,\ y]$	`[27,-41]`
16	$[-Y,\ X,\ -X]$	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y\right]$	`[28,-43]`
17	$[X,\ -Y,\ -X]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -y\right]$	`[29,-39]`
18	$[X,\ Y,\ X]$	$\left[\frac{1}{2}-y,\ 0,\ y+\frac{1}{2}\right]$	`[30,-37]`
19	$[-X,\ Y,\ -X]$	$[y,\ 0,\ -y]$	`[31,-40]`
20	$[-X,\ -Y,\ X]$	$\left[y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	`[32,-38]`
21	$[-X,\ X,\ -Y]$	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[33,-45]`
22	$[X,\ X,\ Y]$	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[34,-47]`
23	$[-X,\ -X,\ Y]$	$[-y,\ y,\ 0]$	`[35,-46]`
24	$[X,\ -X,\ -Y]$	$\left[y+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[36,-48]`

Table 42: Wyckoff bond: `24b@24i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ X,\ X]$	$\left[\frac{1}{2},\ y,\ y+\frac{1}{2}\right]$	`[1,18]`
2	$[0,\ -X,\ X]$	$\left[0,\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	`[2,20]`
3	$[0,\ X,\ -X]$	$[0,\ y,\ -y]$	`[3,17]`
4	$[0,\ -X,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -y\right]$	`[4,19]`
5	$[X,\ 0,\ X]$	$\left[y+\frac{1}{2},\ \frac{1}{2},\ y\right]$	`[5,15]`
6	$[X,\ 0,\ -X]$	$\left[y+\frac{1}{2},\ 0,\ \frac{1}{2}-y\right]$	`[6,13]`
7	$[-X,\ 0,\ X]$	$[-y,\ 0,\ y]$	`[7,16]`
8	$[-X,\ 0,\ -X]$	$\left[-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	`[8,14]`
9	$[X,\ X,\ 0]$	$\left[y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,21]`
10	$[-X,\ X,\ 0]$	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ 0\right]$	`[10,23]`
11	$[X,\ -X,\ 0]$	$[y,\ -y,\ 0]$	`[11,22]`
12	$[-X,\ -X,\ 0]$	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{2}\right]$	`[12,24]`
13	$[0,\ -X,\ -X]$	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-y\right]$	`[25,42]`
14	$[0,\ X,\ -X]$	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[26,44]`
15	$[0,\ -X,\ X]$	$[0,\ -y,\ y]$	`[27,41]`
16	$[0,\ X,\ X]$	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y\right]$	`[28,43]`
17	$[-X,\ 0,\ -X]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -y\right]$	`[29,39]`
18	$[-X,\ 0,\ X]$	$\left[\frac{1}{2}-y,\ 0,\ y+\frac{1}{2}\right]$	`[30,37]`
19	$[X,\ 0,\ -X]$	$[y,\ 0,\ -y]$	`[31,40]`
20	$[X,\ 0,\ X]$	$\left[y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	`[32,38]`
21	$[-X,\ -X,\ 0]$	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[33,45]`
22	$[X,\ -X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[34,47]`
23	$[-X,\ X,\ 0]$	$[-y,\ y,\ 0]$	`[35,46]`
24	$[X,\ X,\ 0]$	$\left[y+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[36,48]`

Table 43: Wyckoff bond: `48c@24i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, y+\frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}-y, y+\frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -y]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}-y, -y\right]$	`[4]`
5	$[Z, X, Y]$	$\left[y+\frac{1}{2}, \frac{1}{2}, y\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[y+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	`[6]`
7	$[-Z, -X, Y]$	$[-y, 0, y]$	`[7]`
8	$[-Z, X, -Y]$	$\left[-y, \frac{1}{2}, \frac{1}{2}-y\right]$	`[8]`
9	$[Y, Z, X]$	$\left[y, y+\frac{1}{2}, \frac{1}{2}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, 0\right]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -y, 0]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, -y, \frac{1}{2}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[-y, \frac{1}{2}, \frac{1}{2}-y\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[y+\frac{1}{2}, \frac{1}{2}, y\right]$	`[15]`
16	$[-Y, X, Z]$	$[-y, 0, y]$	`[16]`
17	$[X, Z, -Y]$	$[0, y, -y]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, y, y+\frac{1}{2}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}-y, -y\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[0, \frac{1}{2}-y, y+\frac{1}{2}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[y, y+\frac{1}{2}, \frac{1}{2}\right]$	`[21]`
22	$[Z, -Y, X]$	$[y, -y, 0]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, 0\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-y, -y, \frac{1}{2}\right]$	`[24]`
25	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}-y\right]$	`[25]`
26	$[X, Y, -Z]$	$\left[0, y+\frac{1}{2}, \frac{1}{2}-y\right]$	`[26]`
27	$[X, -Y, Z]$	$[0, -y, y]$	`[27]`
28	$[-X, Y, Z]$	$\left[\frac{1}{2}, y+\frac{1}{2}, y\right]$	`[28]`
29	$[-Z, -X, -Y]$	$\left[\frac{1}{2}-y, \frac{1}{2}, -y\right]$	`[29]`
30	$[-Z, X, Y]$	$\left[\frac{1}{2}-y, 0, y+\frac{1}{2}\right]$	`[30]`
31	$[Z, X, -Y]$	$[y, 0, -y]$	`[31]`
32	$[Z, -X, Y]$	$\left[y, \frac{1}{2}, y+\frac{1}{2}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$\left[-y, \frac{1}{2}-y, \frac{1}{2}\right]$	`[33]`
34	$[Y, -Z, X]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	`[34]`
35	$[-Y, Z, X]$	$[-y, y, 0]$	`[35]`
36	$[Y, Z, -X]$	$\left[y+\frac{1}{2}, y, \frac{1}{2}\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}-y, 0, y+\frac{1}{2}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[y, \frac{1}{2}, y+\frac{1}{2}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}, -y\right]$	`[39]`
40	$[Y, -X, -Z]$	$[y, 0, -y]$	`[40]`
41	$[-X, -Z, Y]$	$[0, -y, y]$	`[41]`
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, -y, \frac{1}{2}-y\right]$	`[42]`
43	$[X, Z, Y]$	$\left[\frac{1}{2}, y+\frac{1}{2}, y\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[0, y+\frac{1}{2}, \frac{1}{2}-y\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[-y, \frac{1}{2}-y, \frac{1}{2}\right]$	`[45]`
46	$[-Z, Y, -X]$	$[-y, y, 0]$	`[46]`

continued ...

Table 43

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[47]
48	$[Z, Y, X]$	$\left[y+\frac{1}{2}, y, \frac{1}{2}\right]$	[48]

* Wyckoff site: 24j, site symmetry: ..2

Table 44: Wyckoff bond: 24a@24j, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{2}, y, -y\right]$	[1,-19]
2	$[-Y, -X, X]$	$\left[0, \frac{1}{2}-y, -y\right]$	[2,-17]
3	$[-Y, X, -X]$	$\left[0, y, y+\frac{1}{2}\right]$	[3,-20]
4	$[Y, -X, -X]$	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[4,-18]
5	$[X, Y, X]$	$\left[-y, \frac{1}{2}, y\right]$	[5,-14]
6	$[X, -Y, -X]$	$\left[-y, 0, \frac{1}{2}-y\right]$	[6,-16]
7	$[-X, -Y, X]$	$\left[y+\frac{1}{2}, 0, y\right]$	[7,-13]
8	$[-X, Y, -X]$	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	[8,-15]
9	$[X, X, Y]$	$\left[y, -y, \frac{1}{2}\right]$	[9,-24]
10	$[-X, X, -Y]$	$\left[\frac{1}{2}-y, -y, 0\right]$	[10,-22]
11	$[X, -X, -Y]$	$\left[y, y+\frac{1}{2}, 0\right]$	[11,-23]
12	$[-X, -X, Y]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	[12,-21]
13	$[-Y, -X, -X]$	$\left[\frac{1}{2}, -y, y\right]$	[25,-43]
14	$[Y, X, -X]$	$\left[0, y+\frac{1}{2}, y\right]$	[26,-41]
15	$[Y, -X, X]$	$\left[0, -y, \frac{1}{2}-y\right]$	[27,-44]
16	$[-Y, X, X]$	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[28,-42]
17	$[-X, -Y, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[29,-38]
18	$[-X, Y, X]$	$\left[y, 0, y+\frac{1}{2}\right]$	[30,-40]
19	$[X, Y, -X]$	$\left[\frac{1}{2}-y, 0, -y\right]$	[31,-37]
20	$[X, -Y, X]$	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	[32,-39]
21	$[-X, -X, -Y]$	$\left[-y, y, \frac{1}{2}\right]$	[33,-48]
22	$[X, -X, Y]$	$\left[y+\frac{1}{2}, y, 0\right]$	[34,-46]
23	$[-X, X, Y]$	$\left[-y, \frac{1}{2}-y, 0\right]$	[35,-47]
24	$[X, X, -Y]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[36,-45]

Table 45: Wyckoff bond: 24b@24j, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{2}, y, -y\right]$	[1,19]
2	$[0, -X, -X]$	$\left[0, \frac{1}{2}-y, -y\right]$	[2,17]
3	$[0, X, X]$	$\left[0, y, y+\frac{1}{2}\right]$	[3,20]
4	$[0, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[4,18]
5	$[-X, 0, X]$	$\left[-y, \frac{1}{2}, y\right]$	[5,14]
6	$[-X, 0, -X]$	$\left[-y, 0, \frac{1}{2}-y\right]$	[6,16]
7	$[X, 0, X]$	$\left[y+\frac{1}{2}, 0, y\right]$	[7,13]
8	$[X, 0, -X]$	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	[8,15]
9	$[X, -X, 0]$	$\left[y, -y, \frac{1}{2}\right]$	[9,24]

continued ...

Table 45

No.	vector	center	mapping
10	$[-X, -X, 0]$	$\left[\frac{1}{2}-y, -y, 0\right]$	[10,22]
11	$[X, X, 0]$	$\left[y, y+\frac{1}{2}, 0\right]$	[11,23]
12	$[-X, X, 0]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	[12,21]
13	$[0, -X, X]$	$\left[\frac{1}{2}, -y, y\right]$	[25,43]
14	$[0, X, X]$	$\left[0, y+\frac{1}{2}, y\right]$	[26,41]
15	$[0, -X, -X]$	$\left[0, -y, \frac{1}{2}-y\right]$	[27,44]
16	$[0, X, -X]$	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[28,42]
17	$[X, 0, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[29,38]
18	$[X, 0, X]$	$\left[y, 0, y+\frac{1}{2}\right]$	[30,40]
19	$[-X, 0, -X]$	$\left[\frac{1}{2}-y, 0, -y\right]$	[31,37]
20	$[-X, 0, X]$	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	[32,39]
21	$[-X, X, 0]$	$\left[-y, y, \frac{1}{2}\right]$	[33,48]
22	$[X, X, 0]$	$\left[y+\frac{1}{2}, y, 0\right]$	[34,46]
23	$[-X, -X, 0]$	$\left[-y, \frac{1}{2}-y, 0\right]$	[35,47]
24	$[X, -X, 0]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[36,45]

Table 46: Wyckoff bond: 48c@24j, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, -y\right]$	[1]
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}-y, -y\right]$	[2]
3	$[-X, Y, -Z]$	$\left[0, y, y+\frac{1}{2}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[4]
5	$[Z, X, Y]$	$\left[-y, \frac{1}{2}, y\right]$	[5]
6	$[Z, -X, -Y]$	$\left[-y, 0, \frac{1}{2}-y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[y+\frac{1}{2}, 0, y\right]$	[7]
8	$[-Z, X, -Y]$	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	[8]
9	$[Y, Z, X]$	$\left[y, -y, \frac{1}{2}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-y, -y, 0\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y, y+\frac{1}{2}, 0\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, 0, y\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[-y, \frac{1}{2}, y\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	[15]
16	$[-Y, X, Z]$	$\left[-y, 0, \frac{1}{2}-y\right]$	[16]
17	$[X, Z, -Y]$	$\left[0, \frac{1}{2}-y, -y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, y, -y\right]$	[19]
20	$[X, -Z, Y]$	$\left[0, y, y+\frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}-y, -y, 0\right]$	[22]
23	$[-Z, Y, X]$	$\left[y, y+\frac{1}{2}, 0\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[y, -y, \frac{1}{2}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, y\right]$	[25]
26	$[X, Y, -Z]$	$\left[0, y+\frac{1}{2}, y\right]$	[26]

continued ...

Table 46

No.	vector	center	mapping
27	$[X, -Y, Z]$	$\left[0, -y, \frac{1}{2}-y\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[y, \frac{1}{2}, -y\right]$	[29]
30	$[-Z, X, Y]$	$\left[y, 0, y+\frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{2}-y, 0, -y\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[-y, y, \frac{1}{2}\right]$	[33]
34	$[Y, -Z, X]$	$\left[y+\frac{1}{2}, y, 0\right]$	[34]
35	$[-Y, Z, X]$	$\left[-y, \frac{1}{2}-y, 0\right]$	[35]
36	$[Y, Z, -X]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}-y, 0, -y\right]$	[37]
38	$[Y, X, Z]$	$\left[y, \frac{1}{2}, -y\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y, 0, y+\frac{1}{2}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[0, y+\frac{1}{2}, y\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{2}, -y, y\right]$	[43]
44	$[-X, Z, -Y]$	$\left[0, -y, \frac{1}{2}-y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[y+\frac{1}{2}, y, 0\right]$	[46]
47	$[Z, -Y, -X]$	$\left[-y, \frac{1}{2}-y, 0\right]$	[47]
48	$[Z, Y, X]$	$\left[-y, y, \frac{1}{2}\right]$	[48]

* Wyckoff site: 24k, site symmetry: ..m

Table 47: Wyckoff bond: 24a@24k, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, z]$	[1,38]
2	$[-X, -X, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	[2,37]
3	$[-X, X, -Y]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-z\right]$	[3,39]
4	$[X, -X, -Y]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[4,40]
5	$[Y, X, X]$	$[z, x, x]$	[5,48]
6	$[Y, -X, -X]$	$\left[z, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[6,47]
7	$[-Y, -X, X]$	$\left[\frac{1}{2}-z, \frac{1}{2}-x, x\right]$	[7,45]
8	$[-Y, X, -X]$	$\left[\frac{1}{2}-z, x, \frac{1}{2}-x\right]$	[8,46]
9	$[X, Y, X]$	$[x, z, x]$	[9,43]
10	$[-X, Y, -X]$	$\left[\frac{1}{2}-x, z, \frac{1}{2}-x\right]$	[10,44]
11	$[X, -Y, -X]$	$\left[x, \frac{1}{2}-z, \frac{1}{2}-x\right]$	[11,42]
12	$[-X, -Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-z, x\right]$	[12,41]
13	$[X, X, -Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	[13,26]
14	$[-X, -X, -Y]$	$[-x, -x, -z]$	[14,25]
15	$[X, -X, Y]$	$\left[x+\frac{1}{2}, -x, z+\frac{1}{2}\right]$	[15,27]
16	$[-X, X, Y]$	$\left[-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[16,28]
17	$[X, Y, -X]$	$\left[x+\frac{1}{2}, z+\frac{1}{2}, -x\right]$	[17,36]
18	$[-X, Y, X]$	$\left[-x, z+\frac{1}{2}, x+\frac{1}{2}\right]$	[18,35]
19	$[-X, -Y, -X]$	$[-x, -z, -x]$	[19,33]

continued ...

Table 47

No.	vector	center	mapping
20	$[X, -Y, X]$	$\left[x+\frac{1}{2}, -z, x+\frac{1}{2}\right]$	[20,34]
21	$[Y, X, -X]$	$\left[z+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[21,31]
22	$[Y, -X, X]$	$\left[z+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[22,32]
23	$[-Y, X, X]$	$\left[-z, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[23,30]
24	$[-Y, -X, -X]$	$[-z, -x, -x]$	[24,29]

Table 48: Wyckoff bond: 24b@24k, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	[1,-38]
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	[2,-37]
3	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-z\right]$	[3,-39]
4	$[X, X, 0]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[4,-40]
5	$[0, X, -X]$	$[z, x, x]$	[5,-48]
6	$[0, -X, X]$	$\left[z, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[6,-47]
7	$[0, -X, -X]$	$\left[\frac{1}{2}-z, \frac{1}{2}-x, x\right]$	[7,-45]
8	$[0, X, X]$	$\left[\frac{1}{2}-z, x, \frac{1}{2}-x\right]$	[8,-46]
9	$[-X, 0, X]$	$[x, z, x]$	[9,-43]
10	$[X, 0, -X]$	$\left[\frac{1}{2}-x, z, \frac{1}{2}-x\right]$	[10,-44]
11	$[-X, 0, -X]$	$\left[x, \frac{1}{2}-z, \frac{1}{2}-x\right]$	[11,-42]
12	$[X, 0, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-z, x\right]$	[12,-41]
13	$[-X, X, 0]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	[13,-26]
14	$[X, -X, 0]$	$[-x, -x, -z]$	[14,-25]
15	$[-X, -X, 0]$	$\left[x+\frac{1}{2}, -x, z+\frac{1}{2}\right]$	[15,-27]
16	$[X, X, 0]$	$\left[-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[16,-28]
17	$[X, 0, X]$	$\left[x+\frac{1}{2}, z+\frac{1}{2}, -x\right]$	[17,-36]
18	$[-X, 0, -X]$	$\left[-x, z+\frac{1}{2}, x+\frac{1}{2}\right]$	[18,-35]
19	$[-X, 0, X]$	$[-x, -z, -x]$	[19,-33]
20	$[X, 0, -X]$	$\left[x+\frac{1}{2}, -z, x+\frac{1}{2}\right]$	[20,-34]
21	$[0, -X, -X]$	$\left[z+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[21,-31]
22	$[0, X, X]$	$\left[z+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[22,-32]
23	$[0, -X, X]$	$\left[-z, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[23,-30]
24	$[0, X, -X]$	$[-z, -x, -x]$	[24,-29]

Table 49: Wyckoff bond: 48c@24k, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[4]
5	$[Z, X, Y]$	$[z, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[z, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-z, \frac{1}{2}-x, x\right]$	[7]

continued ...

Table 49

No.	vector	center	mapping
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-x\right]$	[8]
9	$[Y,\ Z,\ X]$	$[x,\ z,\ x]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[\frac{1}{2}-x,\ z,\ \frac{1}{2}-x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ x\right]$	[12]
13	$[Y,\ X,\ -Z]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[13]
14	$[-Y,\ -X,\ -Z]$	$[-x,\ -x,\ -z]$	[14]
15	$[Y,\ -X,\ Z]$	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$[-Y,\ X,\ Z]$	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]
17	$[X,\ Z,\ -Y]$	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[17]
18	$[-X,\ Z,\ Y]$	$\left[-x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[18]
19	$[-X,\ -Z,\ -Y]$	$[-x,\ -z,\ -x]$	[19]
20	$[X,\ -Z,\ Y]$	$\left[x+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[20]
21	$[Z,\ Y,\ -X]$	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[21]
22	$[Z,\ -Y,\ X]$	$\left[z+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[22]
23	$[-Z,\ Y,\ X]$	$\left[-z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23]
24	$[-Z,\ -Y,\ -X]$	$[-z,\ -x,\ -x]$	[24]
25	$[-X,\ -Y,\ -Z]$	$[-x,\ -x,\ -z]$	[25]
26	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[26]
27	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[27]
28	$[-X,\ Y,\ Z]$	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[28]
29	$[-Z,\ -X,\ -Y]$	$[-z,\ -x,\ -x]$	[29]
30	$[-Z,\ X,\ Y]$	$\left[-z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[30]
31	$[Z,\ X,\ -Y]$	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[31]
32	$[Z,\ -X,\ Y]$	$\left[z+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[32]
33	$[-Y,\ -Z,\ -X]$	$[-x,\ -z,\ -x]$	[33]
34	$[Y,\ -Z,\ X]$	$\left[x+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[34]
35	$[-Y,\ Z,\ X]$	$\left[-x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[35]
36	$[Y,\ Z,\ -X]$	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[36]
37	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[37]
38	$[Y,\ X,\ Z]$	$[x,\ x,\ z]$	[38]
39	$[-Y,\ X,\ -Z]$	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-z\right]$	[39]
40	$[Y,\ -X,\ -Z]$	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[40]
41	$[-X,\ -Z,\ Y]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ x\right]$	[41]
42	$[X,\ -Z,\ -Y]$	$\left[x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[42]
43	$[X,\ Z,\ Y]$	$[x,\ z,\ x]$	[43]
44	$[-X,\ Z,\ -Y]$	$\left[\frac{1}{2}-x,\ z,\ \frac{1}{2}-x\right]$	[44]
45	$[-Z,\ -Y,\ X]$	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ x\right]$	[45]
46	$[-Z,\ Y,\ -X]$	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-x\right]$	[46]
47	$[Z,\ -Y,\ -X]$	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[47]
48	$[Z,\ Y,\ X]$	$[z,\ x,\ x]$	[48]

* Wyckoff site: 481, site symmetry: 1

Table 50: Wyckoff bond: `48a@48l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[4]
5	$[Z,\ X,\ Y]$	$[z,\ x,\ y]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[8]
9	$[Y,\ Z,\ X]$	$[y,\ z,\ x]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[12]
13	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[13]
14	$[-Y,\ -X,\ -Z]$	$[-y,\ -x,\ -z]$	[14]
15	$[Y,\ -X,\ Z]$	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$[-Y,\ X,\ Z]$	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]
17	$[X,\ Z,\ -Y]$	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -y\right]$	[17]
18	$[-X,\ Z,\ Y]$	$\left[-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[18]
19	$[-X,\ -Z,\ -Y]$	$[-x,\ -z,\ -y]$	[19]
20	$[X,\ -Z,\ Y]$	$\left[x+\frac{1}{2},\ -z,\ y+\frac{1}{2}\right]$	[20]
21	$[Z,\ Y,\ -X]$	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ -x\right]$	[21]
22	$[Z,\ -Y,\ X]$	$\left[z+\frac{1}{2},\ -y,\ x+\frac{1}{2}\right]$	[22]
23	$[-Z,\ Y,\ X]$	$\left[-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23]
24	$[-Z,\ -Y,\ -X]$	$[-z,\ -y,\ -x]$	[24]
25	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	[25]
26	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[26]
27	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[27]
28	$[-X,\ Y,\ Z]$	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[28]
29	$[-Z,\ -X,\ -Y]$	$[-z,\ -x,\ -y]$	[29]
30	$[-Z,\ X,\ Y]$	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[30]
31	$[Z,\ X,\ -Y]$	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[31]
32	$[Z,\ -X,\ Y]$	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	[32]
33	$[-Y,\ -Z,\ -X]$	$[-y,\ -z,\ -x]$	[33]
34	$[Y,\ -Z,\ X]$	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[34]
35	$[-Y,\ Z,\ X]$	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[35]
36	$[Y,\ Z,\ -X]$	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[36]
37	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[37]
38	$[Y,\ X,\ Z]$	$[y,\ x,\ z]$	[38]
39	$[-Y,\ X,\ -Z]$	$\left[\frac{1}{2}-y,\ x,\ \frac{1}{2}-z\right]$	[39]
40	$[Y,\ -X,\ -Z]$	$\left[y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[40]
41	$[-X,\ -Z,\ Y]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y\right]$	[41]
42	$[X,\ -Z,\ -Y]$	$\left[x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[42]
43	$[X,\ Z,\ Y]$	$[x,\ z,\ y]$	[43]
44	$[-X,\ Z,\ -Y]$	$\left[\frac{1}{2}-x,\ z,\ \frac{1}{2}-y\right]$	[44]
45	$[-Z,\ -Y,\ X]$	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x\right]$	[45]
46	$[-Z,\ Y,\ -X]$	$\left[\frac{1}{2}-z,\ y,\ \frac{1}{2}-x\right]$	[46]

continued ...

Table 50

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$\left[z, \frac{1}{2} - y, \frac{1}{2} - x\right]$	[47]
48	$[Z, Y, X]$	$[z, y, x]$	[48]